University of Toronto
Combustion Processes (AER315)
Instructor: Rajat Sawanni

Handout #1
September 10, 2020

Chemical Thermodynamics

Outline *In this handout, we will cover the first topic forming the basis of combustion studies, i.e. thermo-chemistry. This topic deals with the thermodynamics involved in combustion reactions. We start with an introduction to stoichiometry followed by a review of property relations and equation of state (EOS) for ideal gases and mixtures. The first law is then used to determine the energy released with the chemical reaction and final temperature of the product. An important quantity termed as the adiabatic flame temperature is defined and evaluated to understand the heat releasing capacity of a given reactant mixture. Subsequently, the second law is used to determine the equilibrium concentrations of different product species.*

Contents

1 Terminology

- **System**: The portion of physical universe under thermodynamic study is referred to as the system
- **Surroundings**: The remaining part of the physical universe is termed as the surrounding
- **Boundary**: The real or imaginary surface separating the system and its surrounding is termed as the boundary. This is where the system and surrounding exchange matter and energy with each other.
- **Open system**: A system which can exchange both matter and energy with its surroundings is termed as an open system
- **Closed system**: A system which can exchange energy but not matter with its surrounding is termed as a closed system
- **Isolated system**: A system which cannot exchange matter or energy with its surroundings is termed as an isolated system
- **Homogeneous system**: When the properties of a system are uniform throughout its boundaries or consist of a single phase, it is termed as a homogeneous system
- **Heterogeneous system**: A heterogeneous system is one which is not uniform throughout and contains two or more phases which are separated from one another by a definite boundary.
- **Extensive property**: Properties which are dependent on the quantity of the substance. Usually denoted by capital letters. Examples are: V for volume, U for total internal energy, H for total enthalpy.
- **Intensive property**: Properties which are independent of the quantity of substance. Usually denoted with lower case letters. Examples are: specific volume v, specific enthalpy h, specific heat at constant pressure c_p. Important exceptions to this lower case conventions is temperature T. Molar based properties will be denoted by an overbar, e.g., $\bar{h}$ for specific enthalpy per unit mole, J/mol. Extensive properties may be related to their intensive counterparts by the amount of substance present:

$$V = m \cdot v \text{ (or } \; n \cdot \bar{v})$$
$$H = m \cdot h \text{ (or } \; n \cdot \bar{h})$$

- **State functions/ variables**: Functions or variables are used to define the system at any given thermodynamic state. These are functions of measurable properties of a system. State functions are only dependent on the state of the system and not the path through which it achieved the given state. Some important state functions/variables are given below:
  - p (pressure) - measure of the force per unit area at a given point due to the change in momentum of colliding molecules.
  - V (volume) - the volume covered inside the boundaries of the system.
  - T (Temperature) - a measure of the average kinetic energy of point like masses inside the system. The higher the velocity of these point masses, the higher the temperature.

- u(internal energy) - refers to the internal energy of the particles in terms of their translation, rotation, vibration and electrodynamics potential.
- $h = u+pV$ (enthalpy) - mathematical state function which is useful in calculating certain path functions. Sometimes, it is also related to the amount of useful heat that can be supplied from a system.
- s (entropy) - the entropy of a system is a measure of the number of possible micro-states that the system can be in for any given macro-state. For example, N molecules of gas occupying a smaller volume would have a smaller entropy as compared to the N molecules occupying larger volume. Since larger volume allows for more possibility of the micro states, it has more entropy. In classical thermodynamics, it is defined in terms of change (ds) so that $ds = \frac{\delta Q_{rev}}{T}$
- $g = h-Ts$ (Gibbs free energy) - a change of this state function represents the spontaneity of the process involved. It talks about spontaneity by drawing a comparison of enthalpy favoured (low h) and entropy favoured (high Ts) states to judge if the process would happen on its own (spontaneously)

- **Path functions**: On the other hand, path functions are quantities only measurable during the change of states and are dependent on the path taken during this process. Common example includes quantities like:
  - Heat (q): When energy is exchanged between thermodynamic systems by thermal interaction, the transfer of energy is called heat. The units of heat are therefore the units of energy, or joules (J). Heat is transferred by conduction, convection, and/or radiation.
  - Work (w): Work is the transfer of energy by any process other than heat. Like heat, the unit measurement for work is joules (J). There are many forms of work, including but not limited to mechanical, electrical, and gravitational work. For our purposes, we are concerned with p-V work, which is the work done in an enclosed chemical system.
- **Thermal Equilibrium**: A system is said to be in thermal equilibrium with its surroundings when they are at the same temperature and there is no heat exchange between the system and its surrounding.
- **Mechanical Equilibrium**: A system is said to be in mechanical equilibrium when the net force and net moment acting on the system is zero.
- **Chemical Equilibrium**: A system is said to be in chemical equilibrium when it has no tendency to undergo a spontaneous change in its chemical composition, no matter how slow. The reactions might still continue but the rate of forward and backward reactions is same.
- **Thermodynamic Equilibrium**: When all three kinds of equilibrium are satisfied, the system is said to be in thermodynamic equilibrium.

2 Scope

The foundations for thermodynamics have been laid over 2 centuries ago with the work of Carnot and Thomson. The desire to explore this branch was then fuelled by the motivation to build efficient engines and heat cycles. Fast forward to now, thermodynamics has branched off to many disciplines, each using a different fundamental model as a theoretical or experimental basis, or applying the principles to varying types of systems. The earliest developments are now termed as *'classical thermodynamics'* which deals with the macroscopic properties of systems in equilibrium. More recently, with the development of atomic models, *statistical thermodynamics* has emerged supplementing classical thermodynamics with an interpretation of the microscopic interactions between individual particles or quantum-mechanical states. This field relates the microscopic properties of individual atoms and molecules to the macroscopic (bulk properties of materials that can be observed on the human scale), thereby explaining classical thermodynamics as a natural result of statistics, classical mechanics, and quantum theory at the microscopic level. In terms of rigour, statistical thermodynamics offers a more detailed view of the system and lets us analyse complex microscopic environments using the laws of thermodynamics. However, for a wide range of engineering applications, classical thermodynamics not only provides a considerably more direct approach for analysis and design but also requires far fewer mathematical complications. For these reasons, the macroscopic viewpoint is the one adopted in this course. That being said, the statistical approach will still be used to provide theoretical understanding of certain concepts.

For the study of combustion systems in macro-scale applications, the *'chemical thermodynamics'* branch (or *thermo-chemistry*) provides the most apt environment for analysis of physical and chemical systems. The approach to be taken here is that of irreversible (non-equilibrium) thermodynamics rather than the classical. It is useful to understand the major difference between the two and the choice of non-equilibrium approach for combustion systems. First of all, classical thermodynamics only deals with system in equilibrium and therefore any process must be analysed as a succession of states of thermodynamic equilibrium, i.e. taking place infinitely slowly. In the real world, we must deal with irreversible processes and to do that in any detail, the notions of classical thermodynamics must be supplemented. For systems in *mechanical or thermal non-equilibrium*, the usual procedure is to divide the system into a large number of subsystems that are of infinitesimal size relative to the original system, but still macroscopic relative to the molecular structure of the medium. By assuming that each subsystem is in local equilibrium internally (but not necessarily with its surrounding subsystems), we can apply equilibrium thermodynamics and concept of state variable to the subsystems. This assumption of applying equilibrium state variables puts some interesting constraints on the system as all the extensive state variables are now a summation over these smaller subsystems. For example, it is assumed that the global entropy of the system can be found by simple spatial integration of the local entropy density; this means that spatial structure cannot contribute as it properly should to the global entropy assessment for the system. This approach assumes spatial and temporal continuity and even differentiability of locally defined intensive variables such as temperature and internal energy density. When we come to the description of *chemical non-equilibrium*, the procedure is somewhat different. We shall assume that the system/subsystem is in mechanical and thermal equilibrium and that it is homogeneous in space. Given these conditions, we can now specify the composition of constituent chemical species, and other state variables to the system/subsystem.

3 Equations of State (EOS)

Provides a relationship among the thermodynamic state variables: pressure, temperature, and volume. For Ideal-gases: inter-molecular forces and volume of molecules are ignored and the EOS is given by:

$$pV = n_{tot}R_uT = \sum_i n_iR_uT \tag{1a}$$

$$pV = m_{tot}RT = \sum_i m_iRT \tag{1b}$$

$$pv = RT \tag{1c}$$

$$p = \rho RT \tag{1d}$$

where

$$R = \frac{R_u}{M_{mix}} \tag{2a}$$

$$R_u = 8.314\frac{kJ}{kmol.K} \tag{2b}$$

4 Calorific Equations of State

Relates enthalpy and internal energy to pressure, temperature and volume.

$$u = u(T, v) \tag{3a}$$

$$h = h(T, p) \tag{3b}$$

Differentiating (3a) and (3b):

$$\mathrm{d}u = \left(\frac{\partial u}{\partial T}\right)_v \mathrm{d}T + \left(\frac{\partial u}{\partial v}\right)_T \mathrm{d}v \tag{4a}$$

$$\mathrm{d}h = \left(\frac{\partial h}{\partial T}\right)_p \mathrm{d}T + \left(\frac{\partial h}{\partial p}\right)_T \mathrm{d}p \tag{4b}$$

In (4a) and (4b),

$$c_v = \left(\frac{\partial u}{\partial T}\right)_v \tag{5a}$$

$$c_p = \left(\frac{\partial h}{\partial T}\right)_p \tag{5b}$$

For an ideal gas:

$$(\partial u/\partial v)_T = 0 \tag{6a}$$

$$(\partial h/\partial p)_T = 0 \tag{6b}$$

Integrating (4a), (4b) and substituting (6a), (6b) for an ideal gas:

$$u(T) - u_{\mathrm{ref}} = \int_{T_{\mathrm{ref}}}^{T} c_v \mathrm{d}T \tag{7a}$$

$$h(T) - h_{\mathrm{ref}} = \int_{T_{\mathrm{ref}}}^{T} c_p \mathrm{d}T \tag{7b}$$

The LHS of (7a) is also termed as the sensible enthalpy change. The reference state is hereby defined using the concept of enthalpy of formation.

4.1 Absolute (or Standardized) Enthalpy and Enthalpy of Formation

For any species, we can define an absolute enthalpy that is the sum of an enthalpy of formation and sensible enthalpy change:

$$\underbrace{\bar{h}_i(T)}_{\substack{\text{absolute enthalpy}\\ \text{at temperature T}}} = \underbrace{\bar{h}^o_{f,i}(T_{\text{ref}})}_{\substack{\text{enthalpy of}\\ \text{formation at}\\ \text{standard ref}\\ \text{state } (T_{\text{ref}},p^o)}} + \underbrace{\Delta\bar{h}_{s,i}(T)}_{\substack{\text{sensible enthalpy}\\ \text{change in going}\\ \text{from } T_{\text{ref}} \text{ to } T}} \tag{8}$$

where

$$\Delta\bar{h}_{s,i}(T) \equiv \bar{h}_i(T) - \bar{h}_i(T_{\text{ref}}) \tag{9}$$

Now, the standard reference state is defined using:

$$T_{\text{ref}} = 298.15\text{K } (25\ ^o\text{C}) \tag{10a}$$

$$p_{\text{ref}} = p^o = 1\text{atm } (101.325 \text{ kPa}) \tag{10b}$$

Since it is only possible to measure the change in enthalpy, we arbitrarily take the enthalpies of formation to be zero for the elements in their naturally occuring state at the reference temperature and pressure. For example, at 25 ^{o}C and 1 atm, oxygen exists as diatomic molecules; then

$$(\bar{h}^o_{f,\text{O}_2})_{298} = 0 \tag{11}$$

To define the enthalpy of formation of other species at reference conditions, we consider their formation from constituent elements in their naturally occurring states. For example, to form oxygen atoms at the standard state, we requires breaking of a chemical bond. The bond dissociation energy of O_2 at standard state is $498,390$ kJ/kmol. Breaking of the bond creates two O atoms; therefore the enthalpy of formation of atomic oxygen is

$$(\bar{h}^o_{f,\text{O}})_{298} = 249,195 \text{ kJ/kmol} \tag{12}$$

Thus, enthalpies of formation is the net change in enthalpy associated with breaking the chemical bonds of the standard state elements and forming new bonds to create the compound of interest. Other examples include:

$$(\bar{h}^o_{f,\text{N}_2})_{298} = 0 \tag{13a}$$

$$(\bar{h}^o_{f,\text{H}_2})_{298} = 0 \tag{13b}$$

$$(\bar{h}^o_{f,\text{N}})_{298} = 472,629 \text{ kJ/kmol} \tag{13c}$$

$$(\bar{h}^o_{f,\text{H}})_{298} = 217,977 \text{ kJ/kmol} \tag{13d}$$

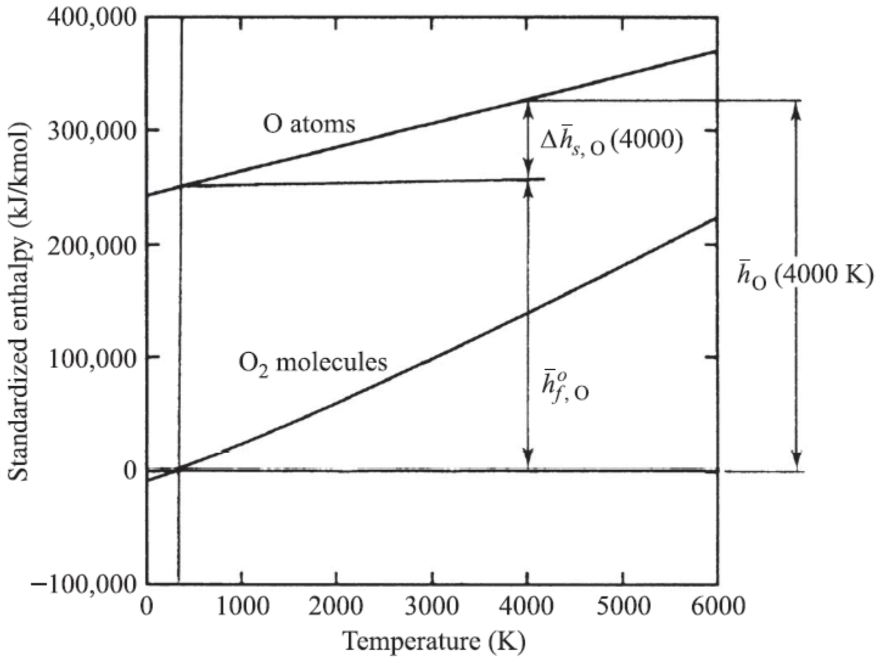

Figure 1: Division of absolute enthalpy to enthalpy of formation and sensible enthalpy

5 Methods for composition specification

5.1 Mole Fraction (χ_i) and Mass Fraction (y_i)

- **Mole Fraction** of a species:

$$\chi_i \equiv \frac{n_i}{\sum\limits_j n_j} = n_i/n_{\mathrm{tot}} \tag{14}$$

- **Mass Fraction** of species:

$$y_i \equiv \frac{m_i}{\sum\limits_j m_j} = m_i/m_{\mathrm{tot}} \tag{15}$$

By definition the sum of all the mole (and mass) fractions must be unity:

$$\sum_i \chi_i = 1 \tag{16a}$$

$$\sum_i y_i = 1 \tag{16b}$$

- **Mean Molecular Mass** of a mixture:

$$\overline{M} = \sum_i \chi_i M_i \tag{17a}$$

$$\overline{M} = \frac{1}{\sum\limits_i (y_i/M_i)} \tag{17b}$$

Relations between χ_i and y_i:

$$y_i = \chi_i M_i/\overline{M} \tag{18a}$$
$$\chi_i = y_i \overline{M}/M_i \tag{18b}$$

- **Partial Pressure** of the i^{th} species is the pressure of this species if it were isolated from the mixture at the same temperature and volume as the mixture. For ideal gases, the mixture pressure is the sum of the constituent partial pressures

$$p = \sum_i p_i \tag{19}$$

The partial pressure can be related to the mixture composition and total pressure as

$$p_i = \chi_i p \tag{20}$$

- For ideal gas mixtures, the **Mixture Enthalpy** is given by:

$$h_{\text{mix}} = \sum_i y_i h_i \tag{21a}$$
$$\bar{h}_{\text{mix}} = \sum_i \chi_i \bar{h}_i \tag{21b}$$

- Similarly, **Mixture Entropy** for ideal gas mixtures is also calculated as a weighted sum of the constituents:

$$s_{\text{mix}}(T,p) = \sum_i y_i s_i(T,p) \tag{22a}$$
$$\bar{s}_{\text{mix}}(T,p) = \sum_i \chi_i \bar{s}_i(T,p) \tag{22b}$$

5.2 Air-Fuel Ratio (A/F) and Fuel Equivalence Ratio (Φ)

- Composition of '**Air**' in general combustion reactions is taken as $\mathbf{O_2 + 3.76N_2}$.
- **Air-Fuel Ratio** is defined as :

$$(A/F) = \frac{mass\ of\ air}{mass\ of\ fuel} \tag{23}$$

- A **Stoichiometric Mixture** is one where the amount of oxidizer is just enough to completely consume the fuel present in the mixture. The Air-Fuel Ratio for a stoichiometric mixture is denoted by $(A/F)_{stoich}$.
- **Fuel Equivalence Ratio (Φ)** is then defined as:

$$\Phi = \frac{(A/F)_{stoich}}{(A/F)_{actual}} \tag{24}$$

- $\Phi = 1$: stoichiometric combustion
- $\Phi < 1$: lean mixture, lean combustion
- $\Phi > 1$: rich mixture, rich combustion

- European convention (and to a certain extent Japanese) is to use Air equivalence ratio, : $\lambda = 1/\Phi$. In certain industries, excess air ratio, excess oxygen, and similar terminologies are also used.
- Other terminologies relating to equivalence ratio are:

$$\% \text{ stoichiometric air} = \frac{100\%}{\Phi} \tag{25a}$$

$$\% \text{ excess air} = \frac{(1-\Phi)}{\Phi} \cdot 100\% \tag{25b}$$

5.3 Mixture Fraction (f)

The conserved scalar concept greatly simplifies the solution of reacting flow problems (i.e., the determination of the fields of velocity, species, and temperature), particularly those involving non-premixed flames. A conserved scalar is any scalar property that is conserved throughout the flowfield. For example, absolute enthalpy would qualify as being a conserved scalar if there are no sinks or sources of thermal energy. Element mass fractions are also conserved scalars. **Mixture fraction (f)** is a conserved scalar, which can be defined in the most general sense as the mass percentage of fuel (both burned and unburned) in a gaseous mixture, i.e.,

$$f \equiv \frac{mass\ of\ material\ having\ its\ origin\ in\ the\ fuel\ stream}{mass\ of\ mixture} \tag{26}$$

When defined this way, an interesting property of mixture fraction is that it has no source term, i.e. it cannot be created or destroyed by chemical reactions. Since fuel reacts with oxidiser in stoichiometric proportions, the percentage mass of material coming from fuel stream remains constant as the reaction progresses. This property has interesting implications which would be noted in later chapters.

6 Zeroth Law of Thermodynamics

The zeroth law of thermodynamics states that if two thermodynamic systems are each in thermal equilibrium with a third one, then they are in thermal equilibrium with each other. Though this may seem like an obvious statement, the zeroth law allows us to quantify the thermal activity of a system using an arbitrary scale of temperature. Lets take an example: Let's say we want to quantify and compare an interesting relation humans experience: love. We would soon run into trouble because this relation is not transitive in the first place, i.e. if A loves B and B loves C, it does not necessarily imply anything about the relation between A and C. Consequently, we cannot quantify it with a number or a relative scale. On the other hand, thermal equilibrium between systems is a transitive relation. Since thermal equilibrium is a transitive property, we can define an arbitrary tagging of different states of thermal equilibrium (i.e. states which are not in thermal equilibrium). Although the labeling may be quite arbitrary, temperature is just such a labeling process which uses the real number system for tagging. The zeroth law justifies the use of suitable thermodynamic systems as thermometers to provide such a labeling, which yield any number of possible empirical temperature scales, and justifies the use of the second law of thermodynamics to provide an absolute, or thermodynamic temperature scale. Such temperature scales bring additional continuity and ordering (i.e., "hot" and "cold") properties to the concept of temperature.

7 First Law of Thermodynamics: Energy Balance

The first law of thermodynamics is essentially a statement of energy conservation in that energy cannot be created or destroyed but can only change from one form to another. In the simplest form, the first law for thermodynamic system states that during any cycle that a system undergoes, the cyclic integral of heat is equal to the cyclic integral of the work.

$$\oint \delta\hat{Q} = \oint \delta\hat{W} \tag{27}$$

where $\oint \delta\hat{Q}$ is the net heat transferred during the cycle, and $\oint \delta\hat{W}$ is the net work dine during the cycle. The first law also implies the existence of a state function represented as the stored energy, E of the system and relates the change of this function to the flow of energy from the surroundings.

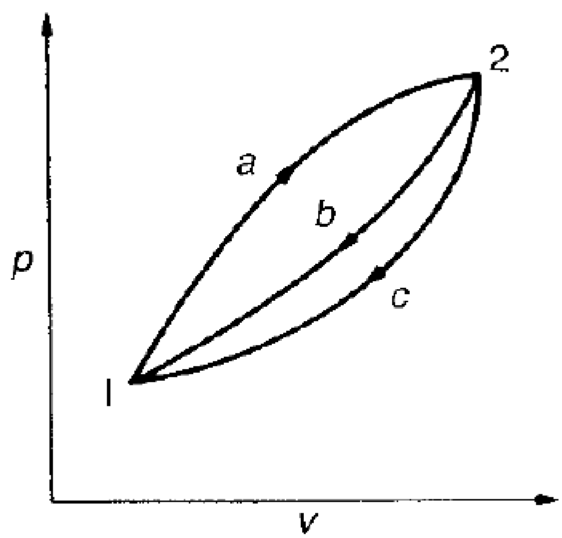

Figure 2: Demonstration of the existence of thermodynamic state variable E

The existence of this thermodynamic property E can be demonstrated as follows from .??: Using (27), we can write the cyclic integral of heat and work by taking the paths a and b as

$$\int_{1a}^{2a} \delta\hat{Q} + \int_{2b}^{1b} \delta\hat{Q} = \int_{1a}^{2a} \delta\hat{W} + \int_{2b}^{1b} \delta\hat{W} \tag{28}$$

We now consider another cycle along the path a and c:

$$\int_{1a}^{2a} \delta\hat{Q} + \int_{2c}^{1c} \delta\hat{Q} = \int_{1a}^{2a} \delta\hat{W} + \int_{2c}^{1c} \delta\hat{W} \tag{29}$$

Subtracting (29) from (28)

$$\int_{2b}^{1b} (\delta\hat{Q} - \delta\hat{W}) = \int_{2c}^{1c} (\delta\hat{Q} - \delta\hat{W}) \tag{30}$$

Since b and c represents arbitrary processes between states 1 and 2, $(\delta\hat{Q} - \delta\hat{W})$ is the same for all processes between 1 and 2. Therefore, we can conclude that $(\delta\hat{Q} - \delta\hat{W})$ is only dependent on the initial and final states and not on the path followed between the two states and is consequently a state function of the thermodynamic system. This property is the energy of the system and is represented by the variable E. We then have:

$$dE \equiv \delta\hat{Q} - \delta\hat{W} \tag{31}$$

The physical significance of the property E is that it represents all the energy of the system in the given state. This energy can be present in many forms, including thermal energy, kinetic energy, potential energy, energy associated with the atomic structure, electrostatic energy etc. It is generally convenient to study the bulk kinetic and potential energy of the system separately and lump all the other forms of energy of the system into a single state variable called the *internal energy*, given by U.

$$E = U + KE_{bulk} + PE_{bulk} \tag{32a}$$

$$dE = dU + d\left(\frac{1}{2}m|v|^2\right) + d(mgz) \tag{32b}$$

7.1 First Law: closed system

For a closed system, energy conservation is expressed for a finite change between two states, 1 and 2, as:

$$\underbrace{{}_1Q_2}_{\substack{\text{Heat added}\\ \text{to the system}}} \quad - \quad \underbrace{{}_1W_2}_{\substack{\text{Work done}\\ \text{by the system}}} \quad = \quad \underbrace{\Delta E_{1-2}}_{\substack{\text{Change in total}\\ \text{energy of system}}} \tag{33}$$

Both ${}_1Q_2$ and ${}_1W_2$ are path functions and occur only at the system boundaries. $\Delta E_{1-2} \equiv E_2 - E_1$ is the change in the total energy of the system, i.e.

$$E = m\{ \underbrace{u}_{\substack{\text{system internal}\\ \text{energy}}} + \underbrace{(1/2)v^2}_{\substack{\text{system' bulk}\\ \text{kinetic energy}}} + \underbrace{gz}_{\substack{\text{system bulk}\\ \text{potential energy}}} \} \tag{34}$$

The system energy is a state variable and does not depend on the path taken. Since this is a closed system, we can write (**??**) on a unit mass basis, or expressed to represent an instant in time:

$${}_1q_2 - {}_1w_2 = \Delta e_{1-2} = e_2 - e_1 \tag{35}$$

$$\underbrace{\dot{Q}}_{\substack{\text{instantaneous}\\ \text{rate of heat}\\ \text{transferred}}} \quad - \quad \underbrace{\dot{W}}_{\substack{\text{instantaneous}\\ \text{rate of work}\\ \text{done}}} \quad = \quad \underbrace{\mathrm{d}E/\mathrm{d}t}_{\substack{\text{instantaneous}\\ \text{change of}\\ \text{system energy}}} \tag{36}$$

7.2 First Law: Control Volume Steady State Steady Flow

Important assumptions:

- The control volume is fixed relative to the coordinate system.
- The properties of the fluid at each point within CV, or on the control surface, do not vary with time.
- There is no overall change in mass of the system so that the rate of incoming mass is balanced by the rate of outgoing mass ($\dot{m}_{in} = \dot{m}_{out} = \dot{m}$).
- Fluid properties are steady and uniform over inlet and outlet areas.
- There is only one inlet and one outlet stream.

Given these assumptions, the steady state, steady flow form of the First Law is given by:

$$\underbrace{\dot{Q}_{cv}}_{\substack{\text{Rate of heat}\\ \text{transferred from}\\ \text{surroundings}}} - \underbrace{\dot{W}_{cv}}_{\substack{\text{Rate of work}\\ \text{done by control}\\ \text{volume excluding}\\ \text{flow work}}} = \underbrace{\dot{m}e_{\mathrm{o}}}_{\substack{\text{Rate of energy}\\ \text{flow out of CV}}} - \underbrace{\dot{m}e_{\mathrm{i}}}_{\substack{\text{Rate of energy}\\ \text{flow into CV}}} + \underbrace{\dot{m}(p_{\mathrm{o}}v_{\mathrm{o}} - p_{\mathrm{i}}v_{\mathrm{i}})}_{\substack{\text{Net rate of work}\\ \text{associated with}\\ \text{pressure forces}}} \tag{37}$$

Specific energy e of the inlet and outlet stream consist of:

$$\underbrace{e}_{\substack{\text{Total}\\ \text{energy per}\\ \text{unit mass}}} = \underbrace{u}_{\substack{\text{Internal}\\ \text{energy per}\\ \text{unit mass}}} + \underbrace{(1/2)v^2}_{\substack{\text{Kinetic}\\ \text{energy per}\\ \text{unit mass}}} + \underbrace{gz}_{\substack{\text{Potential}\\ \text{energy per}\\ \text{unit mass}}} \tag{38}$$

where v = velocity where the stream crosses the CV, z = elevation where stream crosses the CV and g = gravitational acceleration
Enthalpy is given by:

$$h \equiv u + pv = u + p/\rho \tag{39}$$

(37), (38) and (39) yield:

$$\dot{Q}_{cv} - \dot{W}_{cv} = \dot{m}[(h_{\mathrm{o}} - h_{\mathrm{i}}) + \frac{1}{2}(v_{\mathrm{o}}^2 - v_{\mathrm{i}}^2) + g(z_{\mathrm{o}} - z_{\mathrm{i}})] \tag{40}$$

On a mass-specific basis, (40) reads:

$$q_{cv} - w_{cv} = (h_{\mathrm{o}} - h_{\mathrm{i}}) + (1/2)(v_{\mathrm{o}}^2 - v_{\mathrm{i}}^2) + g(z_{\mathrm{o}} - z_{\mathrm{i}}) \tag{41}$$

7.3 Enthalpy of Combustion and Heating Values

A steady flow reactor with no external work is analysed to calculate the enthalpy of combustion and heating values. The reactants are assumed to be stoichiometric mixture at standard state conditions undergoing complete combustion, and products are also assumed to be at standard state conditions. Consequently, for products to exit at the same T as the reactants, heat must be removed. The amount of heat removed is termed as the **enthalpy of combustion** and can be related to the reactant and product absolute enthalpies by applying the steady-flow form of the first law:

$$q_{\mathrm{cv}} = h_{\mathrm{o}} - h_{\mathrm{i}} = h_{\mathrm{prod}} - h_{\mathrm{reac}} \tag{42}$$

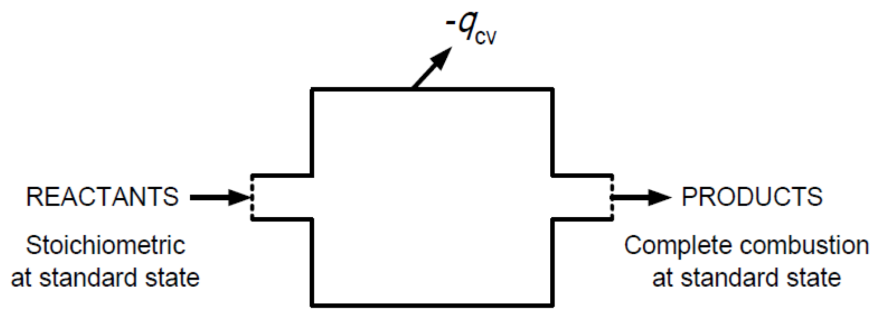

Figure 3: Steady flow reactor with no external work

The enthalpy of reaction, or the enthalpy of combustion, Δh_{R} (per unit mass of mixture), is then given by:

$$\Delta h_{\mathrm{R}} \equiv q_{\mathrm{cv}} = h_{\mathrm{prod}} - h_{\mathrm{reac}} \tag{43}$$

or, in terms of extensive properties,

$$\Delta H_{\mathrm{R}} = H_{\mathrm{prod}} - H_{\mathrm{reac}} \tag{44}$$

For example, at standard state, the reactants enthalpy of a stoichiometric mixture of CH_4 and air, where 1 kmol of fuel reacts, is $-74,831$ kJ. At the same conditions, the combustion products have an absolute enthalpy of $-877,236$ kJ. Consequently, the enthalpy of reaction is given by:

$$\Delta H_{\mathrm{R}} = -877,236 - (-74,831) = -802,405\mathrm{kJ} \tag{45}$$

This value can be adjusted to a per-mass-of-fuel basis:

$$\Delta h_{\mathrm{R}}\Big(\frac{\mathrm{kJ}}{\mathrm{kg_{fuel}}}\Big) = \Delta H_{\mathrm{R}}/M_{\mathrm{fuel}} \tag{46}$$

or,

$$\Delta h_{\mathrm{R}}\Big(\frac{\mathrm{kJ}}{\mathrm{kg_{fuel}}}\Big) = (-802,405/16.043) = -50,016 \tag{47}$$

For a per-unit-mass-of-mixture basis:

$$\Delta h_{\mathrm{R}}\Big(\frac{\mathrm{kJ}}{\mathrm{kg_{mix}}}\Big) = \Delta h_{\mathrm{R}}\Big(\frac{\mathrm{kJ}}{\mathrm{kg_{fuel}}}\Big)\frac{m_{\mathrm{fuel}}}{m_{\mathrm{mix}}} \tag{48}$$

where

$$\frac{m_{\mathrm{fuel}}}{m_{\mathrm{mix}}} = \frac{m_{\mathrm{fuel}}}{m_{\mathrm{air}} + m_{\mathrm{fuel}}} = \frac{1}{(A/F)+1} \tag{49}$$

so that

$$\Delta h_{\mathrm{R}}(\mathrm{kJ/kg_{mix}}) = -50,016/(17.11+1) = -2761.8 \tag{50}$$

It is worthwhile to note that the enthalpy of combustion depends on temperature chosen for its evaluation since enthalpies of reactants and products are temperature dependent. The heat of combustion, Δh_c (known also as the **heating value** or calorific value), is numerically equal to the enthalpy of combustion, but with opposite sign. The heating value is categorised as **Higher Heating Value (HHV)** or **Lower Heating Value (LHV)** depending on the phase of water in product.

- HHV: higher heating value ($H_2O \rightarrow$ liquid)
- LHV: lower heating value ($H_2O \rightarrow$ vapor)

7.4 Adiabatic Flame Temperature (T_{ad})

We will differentiate between two adiabatic flame temperatures:

- for constant-pressure combustion
- for constant-volume combustion

7.4.1 Constant Pressure T_{ad}

If a fuel-air mixture burns adiabatically at constant pressure with no external work done on the system, absolute enthalpy of reactants at the initial state (say, $T_1 = 298$ K, $p = 1$ atm) equals absolute enthalpy of the products at final state ($T = T_{\mathrm{ad}}$, $p = 1$ atm). Consequently, definition of the constant-pressure adiabatic flame temperature is

$$H_{\mathrm{reac}}(T_{\mathrm{i}}, p) = H_{\mathrm{prod}}(T_{\mathrm{ad}}, p) \tag{51}$$

or, on a per-mass-of-mixture basis,

$$h_{\mathrm{reac}}(T_{\mathrm{i}}, p) = h_{\mathrm{prod}}(T_{\mathrm{ad}}, p) \tag{52}$$

Conceptually, adiabatic flame temperature is simple, however, evaluating this quantity requires knowledge of the composition of the combustion products. When we are dealing with constant-pressure combustion systems, such as gas turbine combustors and rocket engines, the appropriate approach would involve constant-pressure T_{ad}.

7.4.2 Constant Volume T_{ad}

If a fuel-air mixture burns adiabatically at constant volume with no external work done on the system, absolute internal energy of reactants at the initial state (say, $T_1 = 298$ K, $p = 1$ atm) equals absolute internal energy of the products at final state ($T = T_{\mathrm{ad}}$, $p = 1$ atm). Consequently, definition of the constant-pressure adiabatic flame temperature is

$$U_{\mathrm{reac}}(T_{\mathrm{init}}, p_{\mathrm{init}}) = U_{\mathrm{prod}}(T_{\mathrm{ad}}, p_{\mathrm{f}}) \tag{53}$$

where U is the absolute (or standardized) internal energy of the mixture. Since most thermodynamic property compilations and calculations provide H (or h) rather than U (or u). So we consider the fact that:

$$H = U + pV \tag{54}$$

Consequently, (53) becomes

$$H_{\mathrm{reac}} - H_{\mathrm{prod}} - V(p_{\mathrm{init}} - p_{\mathrm{f}}) = 0 \tag{55}$$

Now, If we apply the ideal-gas law;

$$p_{\text{init}}V = \sum_{\text{reac}} n_i R_u T_{\text{init}} = n_{\text{reac}} R_u T_{\text{init}} \tag{56a}$$

$$p_{\text{f}}V = \sum_{\text{prod}} n_i R_u T_{\text{ad}} = n_{\text{prod}} R_u T_{\text{ad}} \tag{56b}$$

Substituting back in (55), we get:

$$H_{\text{reac}} - H_{\text{prod}} - R_u(N_{\text{reac}} T_{\text{init}} - N_{\text{prod}} T_{\text{ad}}) = 0 \tag{57}$$

Since

$$m_{\text{mix}}/N_{\text{reac}} \equiv MW_{\text{reac}} \tag{58a}$$

$$m_{\text{mix}}/N_{\text{prod}} \equiv MW_{\text{prod}} \tag{58b}$$

we obtain

$$h_{\text{reac}} - h_{\text{prod}} - R_u\left(\frac{T_{\text{init}}}{MW_{\text{reac}}} - \frac{T_{\text{ad}}}{MW_{\text{prod}}}\right) = 0 \tag{59}$$

When we are dealing with constant-volume combustion systems, such as enclosed explosions or Otto-cycle (idealized thermodynamic cycle for gasoline engine combustion) analysis, the appropriate approach would involve constant-volume T_{ad}.

8 Second Law of Thermodynamics: Entropy

Given the thermodynamic details of a system and the processes it undergoes, the first law helps us in determining the final state from the initial one. However, the application of energy conservation doesn't really distinguish between the processes occurring from state 1 to 2 or from state 2 to 1. This is where the second law comes in and helps us in determining the direction of time, if you will. A general statement of the second law states that the total entropy of an isolated system can never decrease over time, and is constant if and only if all processes are reversible. In other words, isolated systems spontaneously evolve towards thermodynamic equilibrium, the state with maximum entropy. Considering the universe to be an isolated system, we see that the entropy of the universe to be always increasing. This increase in entropy accounts for the irreversibility of natural processes, and the asymmetry between future and past.

For a closed system undergoing a change from one state of thermodynamic equilibrium 1 to another state 2, the entropy change is given by:

$$S_2 - S_1 = \int_1^2 \left(\frac{\delta\hat{Q}}{T}\right)_{rev} \tag{60}$$

where *rev* implies any reversible path between 1 and 2, $\delta\hat{Q}$ is the heat received or added to the system, and T is the absolute temperature of the reservoir that supplies the heat $\delta\hat{Q}$. Since the temperature of the reservoir and the temperature of the system are equal in a reversible process, T is also the temperature of the system. The important point to note here is that since the change in entropy of a substance is path independent. It is also the same for all processes between points 1 and 2. The equation given here enables us to find the entropy change along a reversible path, but once evaluated, the magnitude of entropy change between the two states is the same for all processes. If the same system undergoes an irreversible process from state 1 to state 2, we have;

$$(S_2 - S_1)_{irrev} = S_2 - S_1 > \int_1^2 \left(\frac{\delta \hat{Q}}{T}\right)_{irrev} \tag{61}$$

For irreversible processes, the temperature of the system is undefined under the context of classical thermodynamics. Therefore, we consider non-equilibrium thermodynamics.

8.1 Nonequilibrium Thermodynamics

Given the existence of a state variable called the entropy, S, the change in the entropy, dS in any closed system undergoing any process can be split into two parts:

$$dS = d_eS + d_iS \tag{62}$$

where d_eS is the change in entropy resulting from the interaction between the system and its surroundings (i.e. heat transfer to or from the system) and d_iS is the change in entropy resulting from a process taking place within the system (eg. chemical reactions, mixing of different gases at constant pressure).These differential quantities can also be thought of respectively as the flow of entropy into the system from the surrounding and the production of entropy from irreversible processes within the system. The entropy change d_iS is never negative and is :

$$d_iS = 0 \ (reversible\ process) \tag{63a}$$

$$d_iS > 0 \ (irreversible\ process) \tag{63b}$$

For a closed system undergoing any process, reversible or irreversible, d_eS is given by:

$$d_eS = \frac{\delta \hat{Q}}{T} \tag{64}$$

For a closed system undergoing an irreversible process, (62) can be integrated to give

$$S_2 - S_1 = \int_1^2 dS = \int_1^2 \left(\frac{\delta \hat{Q}}{T}\right) + \int_1^2 d_iS \tag{65}$$

For an irreversible process, $d_iS > 0$ and thus (65) includes the inequality of the classical statement. The latter equation (65) is much more useful than the former (61) in that it replaces the inequality with an equality, and offers a formulation to calculate d_iS

If mass is added to an inert open reversible system, the entropy associated with the extra mass can be considered to be a part of the entropy flow into the system from its surroundings:

$$dS = d_eS = \frac{\delta \hat{Q}}{T} + \sum_j s_j d_e n_j \tag{66}$$

where s_j is the entropy per mole of the j^{th} species added and $d_e n_j$ is the change in number of moles of the j^{th} species flowing into the system. With mass addition, only p-V work and negligible change in bulk potential and kinetic energies, the first law translates to:

$$dU = \delta\hat{Q} - pdV + \sum_j h_j d_e n_j \tag{67}$$

where h_j is the enthalpy per mole associated with the j^{th} species flowing into the system. Now, let us define a new parameter, chemical potential μ_j such that:

$$\mu_j \equiv h_j - Ts_j \tag{68}$$

Dividing (67) by T, we get

$$\frac{\delta\hat{Q}}{T} = \frac{dU}{T} + \frac{pdV}{T} - \frac{1}{T}\sum_j h_j d_e n_j \tag{69}$$

Substituting (69) and (68) to (66) gives

$$dS = d_e S = \frac{1}{T}dU + \frac{p}{T}dV - \frac{1}{T}\sum_j \mu_j d_e n_j \tag{70}$$

for an inert, open, reversible system. The equation (70) is based on the assumption that all the species involved were chemically inert, hence there exists no irreversible processes from chemical reactions within the system. When reactions occur, the mole number of the j^{th} species will change both by mass addition from outside as well as by chemical reactions inside. Therefore,

$$dn_j = d_e n_j + d_i n_j \tag{71}$$

where $d_i n_j$ represents the change in moles of the j^{th} species due to chemical reactions occurring inside the system, while $d_e n_j$ represents the change in mole numbers as a result of the exchange of j^{th} species with the surroundings. If we now consider the j^{th} species to occupy a separate subsystem of the total system, then each subsystem can be regarded as an inert open system. This means that for each species, j;

$$dS_j = \frac{1}{T}dU_j + \frac{p_j}{T}dV - \frac{1}{T}\mu_j dn_j \tag{72}$$

where U_j is the internal energy associated with the j^{th} species. Summing the preceding equations over all subsystems, i.e. for all species and noting that,

$$p = \sum_j p_j \tag{73a}$$

$$S = \sum_j S_j \tag{73b}$$

$$U = \sum_j U_j \tag{73c}$$

we obtain for the total system:

$$dS = \frac{1}{T}dU + \frac{p}{T}dV - \frac{1}{T}\sum_j \mu_j dn_j \tag{74a}$$

$$dU = TdS - pdV + \sum_j \mu_j dn_j \tag{74b}$$

8.2 Free Energy and Equilibrium Constants

It is apparent from the use of definitions of enthalpy, Gibbs free energy and the relation (74b),

$$H = U + pV \tag{75}$$
$$G = H - TS \tag{76}$$

that

$$dH = TdS + Vdp + \sum_j \mu_j dn_j \tag{77}$$
$$dG = -SdT + Vdp + \sum_j \mu_j dn_j \tag{78}$$

By writing the general relation for the total derivative of G with respect to the variables, T, p, and n_j

$$dG = \left(\frac{\partial G}{\partial T}\right)_{p,n_j} dT + \left(\frac{\partial G}{\partial p}\right)_{T,n_j} dp + \sum_j \left(\frac{\partial G}{\partial n_j}\right)_{T,p,n_i(i\neq j)} dn_j \tag{79}$$

Thus, comparing (79) and (78), we get

$$-S = \left(\frac{\partial G}{\partial T}\right)_{p,n_j} \tag{80a}$$
$$V = \left(\frac{\partial G}{\partial p}\right)_{T,n_j} \tag{80b}$$
$$\mu_j = \left(\frac{\partial G}{\partial n_j}\right)_{T,p,n_i(i\neq j)} \tag{80c}$$

or, more generally using equations involving U and H,

$$\mu_j = \left(\frac{\partial G}{\partial n_j}\right)_{T,p,n_i(i\neq j)} = -T\left(\frac{\partial S}{\partial n_j}\right)_{H,p,n_i(i\neq j)} = -T\left(\frac{\partial S}{\partial n_j}\right)_{U,V,n_i(i\neq j)} \tag{81}$$

where (81) can be taken as the definition of μ_j, which is known as the chemical potential and plays an important role in chemical thermodynamics. The chemical potential, an intensive variable, is a state function whose value need not be zero even if the species is not present in the system. There is always a possibility of adding it to the system, in which case the value of G will be altered by the relation given in (81).

The first law of thermodynamics for an open system can be expressed as:

$$dU = \delta\hat{Q} - pdV + d_eU \tag{82}$$

where d_eU is the internal energy carried in by mass addition. Substituting this to (74a) gives:

$$dS = d_eS + d_iS = \frac{1}{T}\underbrace{(\delta\hat{Q} - pdV + d_eU)}_{dU} + \frac{p}{T}dV \underbrace{-\frac{1}{T}\sum_j \mu_j d_en_j - \frac{1}{T}\sum_j \mu_j d_in_j}_{\frac{1}{T}\sum_j \mu_j dn_j} \tag{83}$$

$$dS = \frac{1}{T}\left(\delta\hat{Q} + d_eU - \sum_j \mu_j d_en_j\right) - \frac{1}{T}\sum_j \mu_j d_in_j \tag{84}$$

where the two terms on the RHS correspond to d_eS and d_iS respectively. Thus,

$$d_iS = -\frac{1}{T}\sum_j \mu_j d_i n_j \tag{85}$$

This expression is required by the second law to be either positive (as in the case of an irreversible process) or zero, corresponding to a reversible process. Saying that a process is reversible, in the present context would necessarily mean that the system is in chemical equilibrium at all times, or that the process is one of an infinitely slow succession of states of chemical equilibrium. The condition for chemical equilibrium is therefore given by:

$$d_iS = -\frac{1}{T}\sum_j \mu_j d_i n_j = 0 \tag{86}$$

The quantity $-\sum_j \mu_j d_i n_j$ in chemical literature is known as the *affinity* of the chemical reaction and is commonly denoted by the symbol $a = -\sum_j \mu_j d_i n_j$. At times, it has been used to describe the maximum work obtainable from a chemical process. In a way, it is equivalent to interpreting the Gibbs free energy of the reaction as the driving force for a reaction.

In this context, one may consider that a reaction moves in the direction of decreasing chemical potential, reaching equilibrium only when the potential of the reactants equals that of the products. From (24), we can clearly see that the criteria for equilibrium is also given by:

$$(dG)_{p,T} = 0 \tag{87}$$

Since perfect gases do not exert inter-molecular forces, each component can be analysed by treating them alone in the container. The total mixture Gibbs free enthalpy is then given by:

$$G = \sum_j G_j \tag{88}$$

Using the relation (78) in equilibrium conditions, i.e. with (86), we get:

$$dG = V dp - SdT \tag{89}$$

Assuming the perfect gas relation, for an isothermal process,

$$dG = nR_uTd(\ln p) \tag{90}$$

Hence, one finds:

$$G(T,p) = G^o(T,p^o) + nR_uT\ln(p/p^o) \tag{91}$$

For a given component on a molar basis, this relation can be translated to:

$$\bar{g}_{j,T} = \bar{g}^o_{j,T} + R_uT\ln(p_j/p^o) \tag{92}$$

In reacting systems, similar to the concept of enthalpy of formation, Gibbs function of formation is given by

$$\bar{g}^o_{f,i}(T) \equiv \bar{g}^o_i(T) - \sum_j \nu'_j \bar{g}^o_j(T) \tag{93}$$

ν'_j are the stoichiometric coefficients of the elements required to form one mole of the compound of interest. For example, $\nu'_{\mathrm{O_2}} = 0.5$ and $\nu'_{\mathrm{C}} = 1$ for forming one mole of CO from O_2 and C by the reaction $\mathrm{C} + 0.5\mathrm{O_2} \longrightarrow \mathrm{CO}$. Similar to enthalpies, $\bar{g}^o_{f,i}(T)$ of the naturally occurring elements are assigned values of zero at the reference state. Gibbs function for a mixture of ideal gases is then given by:

$$G_{\mathrm{mix}} = \sum n_j \bar{g}_{j,T} = \sum n_j [\bar{g}^o_{j,T} + R_u T \ln(p_j/p^o)] \tag{94}$$

Applying (87) to the above equation gives:

$$\sum \mathrm{d}n_j [\bar{g}^o_{j,T} + R_u T \ln(p_j/p^o)] + \sum n_j \mathrm{d}[\bar{g}^o_{j,T} + R_u T \ln(p_j/p^o)] = 0 \tag{95}$$

Second term in the last equation is zero, because $\mathrm{d}(\ln p_j) = \mathrm{d}p_j/p_j$ and $\sum \mathrm{d}p_j = 0$ as total pressure is constant. Then,

$$\mathrm{d}G_{\mathrm{mix}} = 0 = \sum \mathrm{d}n_j [\bar{g}^o_{j,T} + R_u T \ln(p_j/p^o)] \tag{96}$$

For a general system, where

$$a\mathrm{A} + b\mathrm{B} + ... \Leftrightarrow e\mathrm{E} + f\mathrm{F} + ... \tag{97}$$

The change in the number of each species is proportional to its stoichiometric coefficient,

$$\begin{aligned} \mathrm{d}n_A &= -\kappa a \\ \mathrm{d}n_B &= -\kappa b \\ . &= . \\ \mathrm{d}n_E &= +\kappa e \\ \mathrm{d}n_F &= +\kappa f \\ . &= . \end{aligned} \tag{98}$$

Replacing (98) to (96). we get;

$$\begin{aligned} &- a[\bar{g}^o_{A,T} + R_u T \ln(p_A/p^o)] \\ &- b[\bar{g}^o_{B,T} + R_u T \ln(p_B/p^o)] - .. \\ &+ e[\bar{g}^o_{E,T} + R_u T \ln(p_E/p^o)] \\ &+ f[\bar{g}^o_{F,T} + R_u T \ln(p_F/p^o)] + .. = 0 \end{aligned} \tag{99}$$

or

$$\begin{aligned} -(e\bar{g}^o_{E,T} + f\bar{g}^o_{F,T} + .. - a\bar{g}^o_{A,T} - b\bar{g}^o_{B,T} - ..) \\ = R_u T \ln \frac{(p_E/p^o)^e \cdot (p_F/p^o)^f \cdot ..}{(p_A/p^o)^a \cdot (p_B/p^o)^b \cdot ..} \end{aligned} \tag{100}$$

Left hand side of (100) is the standard-state Gibbs function change:

$$\Delta G^o_T = (e\bar{g}^o_{E,T} + f\bar{g}^o_{F,T} + .. - a\bar{g}^o_{A,T} - b\bar{g}^o_{B,T} - ..) \tag{101a}$$

$$\Delta G^o_T \equiv (e\bar{g}^o_{f,E} + f\bar{g}^o_{f,F} + .. - a\bar{g}^o_{f,A} - b\bar{g}^o_{f,B} - ..)_T \tag{101b}$$

Argument of the natural logarithm in (100) is defined as the equilibrium constant:

$$K_p = \frac{(p_E/p^o)^e \cdot (p_F/p^o)^f \cdot ..}{(p_A/p^o)^a \cdot (p_B/p^o)^b \cdot ..} \tag{102}$$

Then we have:

$$\Delta G^o_T = -R_u T \ln K_p \tag{103a}$$

$$K_p = \exp[-\Delta G^o_T/(R_u T)] \tag{103b}$$

(103) give a qualitative indication of whether a particular reaction favors products or reactants at equilibrium so that

- **Reactants:** If $\Delta G^o_T > 0 \Rightarrow \ln K_p < 0 \Rightarrow K_p < 1$
- **Products:** If $\Delta G^o_T < 0 \Rightarrow \ln K_p > 0 \Rightarrow K_p > 1$

Similar physical insight can be obtained by considering the definition of ΔG^o_T in terms of enthalpy and entropy changes:

$$\Delta G^o_T = \Delta H^o - T\Delta S^o \tag{104}$$

which can be substituted into (103) to give:

$$K_p = \exp[-\Delta H^o/(R_u T)] \cdot \exp(\Delta S^o/R_u) \tag{105}$$

For $K_p > 1$, which favors products, ΔH^o should be negative (exothermic reaction) and ΔS^o should be positive. These conditions would be ideal for a reaction to go forward as it is favoured by an enthalpy decrease and an entropy increase, thereby making it necessarily spontaneous. On the other hand, a reaction with increased enthalpy and lower entropy would be naturally non-spontaneous and would favour the reactants.